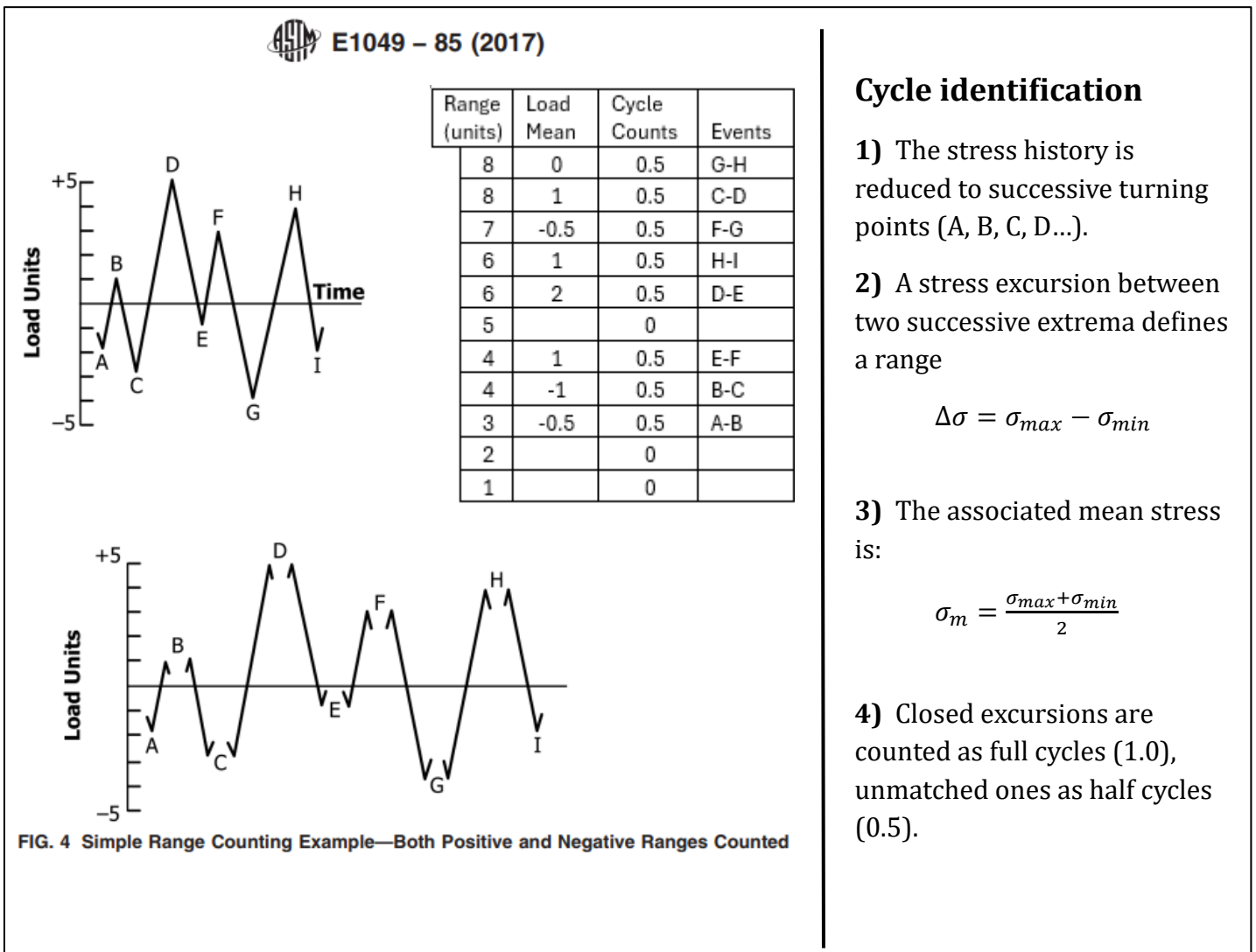


Range-Mean Method

Principle

The range-mean method counts stress excursions between successive extrema while preserving the associated mean stress of each cycle. Each counted cycle is therefore defined by its stress range and its mean value.

ASTM example



Assumptions & Recommended use

Fatigue damage **depends on both stress amplitude and mean stress.**

By preserving the mean value of each cycle, the range-mean method allows **direct integration with Constant Life Diagrams and mean stress correction models.**

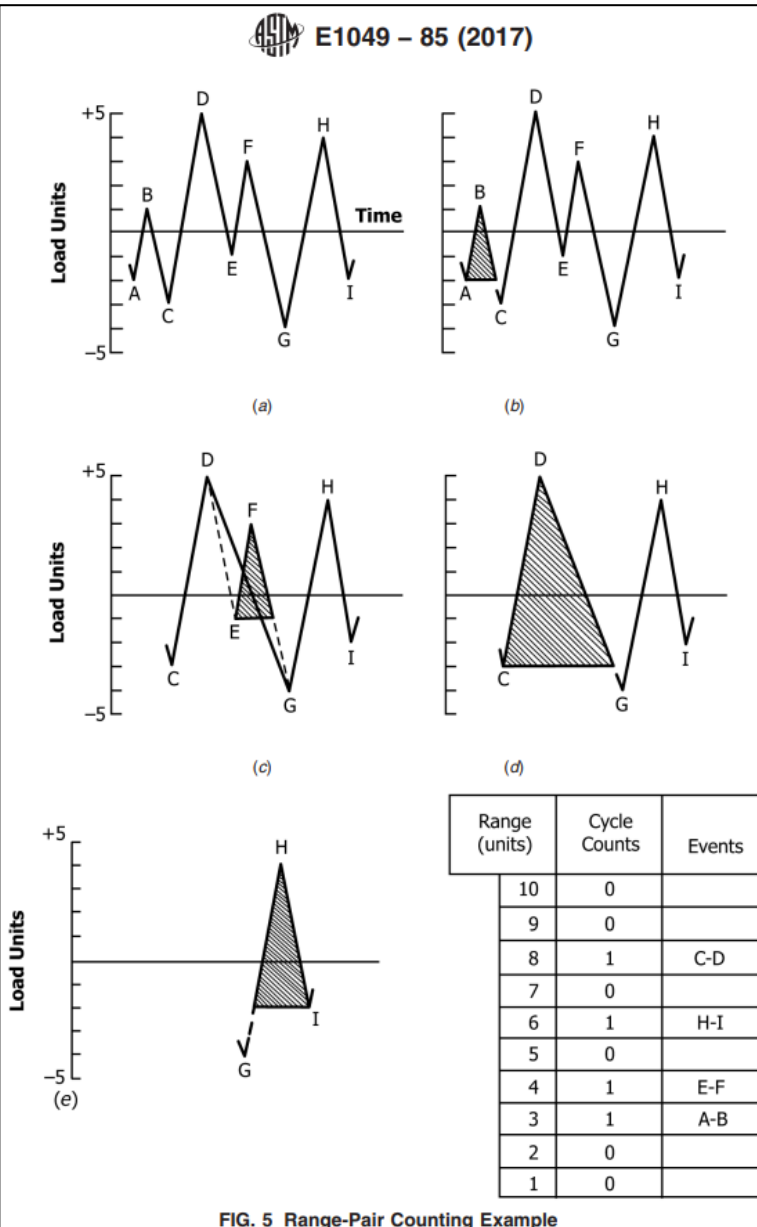
Range-Pair method

Principle

The range-pair method identifies closed stress cycles by comparing adjacent ranges and eliminating smaller inner excursions.

A cycle is counted when a stress range is enclosed by a later range of equal or greater magnitude.

ASTM example



Cycle identification

1) The stress history is reduced to successive turning points (A, B, C, D...).

2) Using the last three non-discarded points, define two consecutive ranges:

$$Y = |p_{n-2} - p_{n-3}|$$

$$X = |p_{n-1} - p_{n-2}|$$

3) Compare absolute values:

- If $X < Y \rightarrow$ continue reading the next point
- If $X \geq Y \rightarrow$
 $\rightarrow$ count Y as one full cycle (1.0)
 $\rightarrow$ remove the two points forming Y
 $\rightarrow$ continue with the shortened sequence

4) End of the sequence

After all comparisons:

- If the last three remaining points do not satisfy the condition $X \geq Y$, the sequence is reversed and the comparison process is repeated.
- If only two turning points remain, the remaining range is counted as a half cycle (0.5).

Assumptions & Recommended use

The method assumes that **fatigue damage is governed by closed stress cycles**.

For load histories with many reversals (e.g., aircraft spectra), the influence of isolated half cycles becomes negligible, making this assumption physically consistent.

Simplified Rain Flow method

Principle

The simplified rainflow method is a variant of rainflow counting intended for repeating load histories. The signal is rearranged so that it starts at the global maximum peak or minimum valley. Cycles are then extracted by comparing adjacent ranges and counting only closed full cycles.

For a repeating history, this method produces only full cycles (no half cycles).

ASTM example

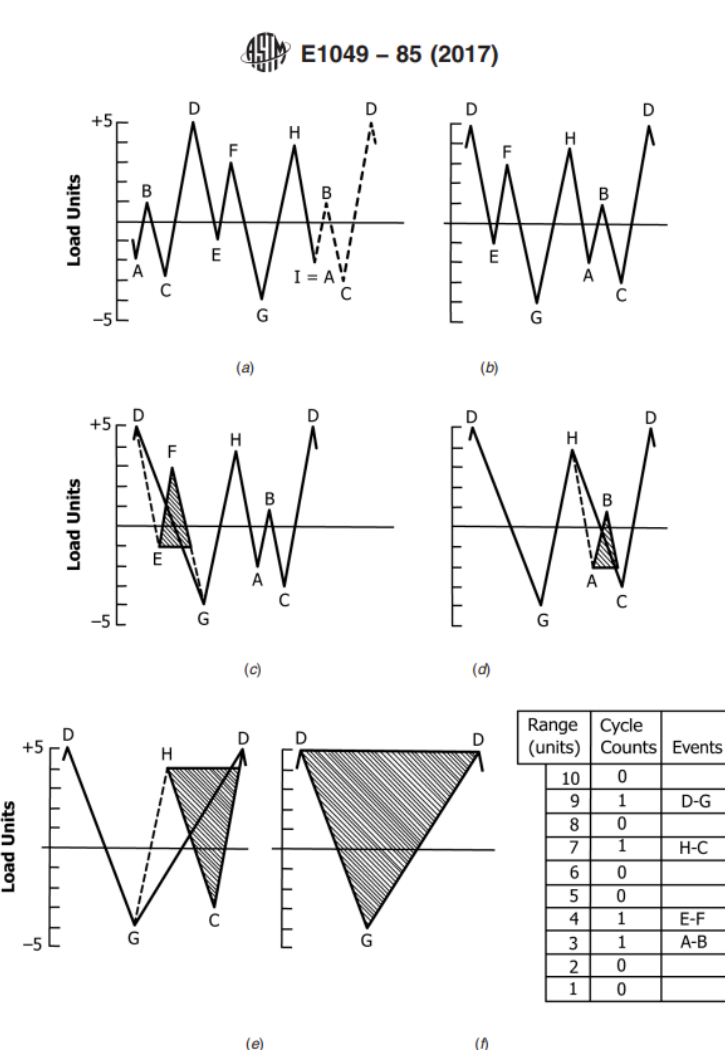


FIG. 7 Example of Simplified Rainflow Counting for a Repeating History

Cycle identification

1) Arrange the signal so that it starts at the maximum peak or minimum valley. This ensures periodicity and eliminates artificial half cycles.

2) Define ranges

Using the three most recent non-discarded turning points:

$$Y = |p_{n-2} - p_{n-3}|$$

$$X = |p_{n-1} - p_{n-2}|$$

3) Compare magnitudes

- If $X < Y \rightarrow$ read next point

- If $X \geq Y \rightarrow$

$\rightarrow$ count Y as one full cycle (1.0)

$\rightarrow$ remove the two points forming Y

$\rightarrow$ continue with shortened sequence

4) End condition

Stop when no further comparison is possible.

Because the history is assumed to repeat, all remaining ranges close naturally, so no half cycles remain.

Assumptions & Recommended use

The method assumes that the load history is cyclically repeated.

Under this assumption, fatigue damage is governed by closed stress-strain hysteresis loops, and only full cycles are relevant.

It provides a consistent cycle count for **periodic spectra without residual half cycles**.

Sources

- ASTM E1049 – 85 (2017), *Standard Practices for Cycle Counting in Fatigue Analysis*